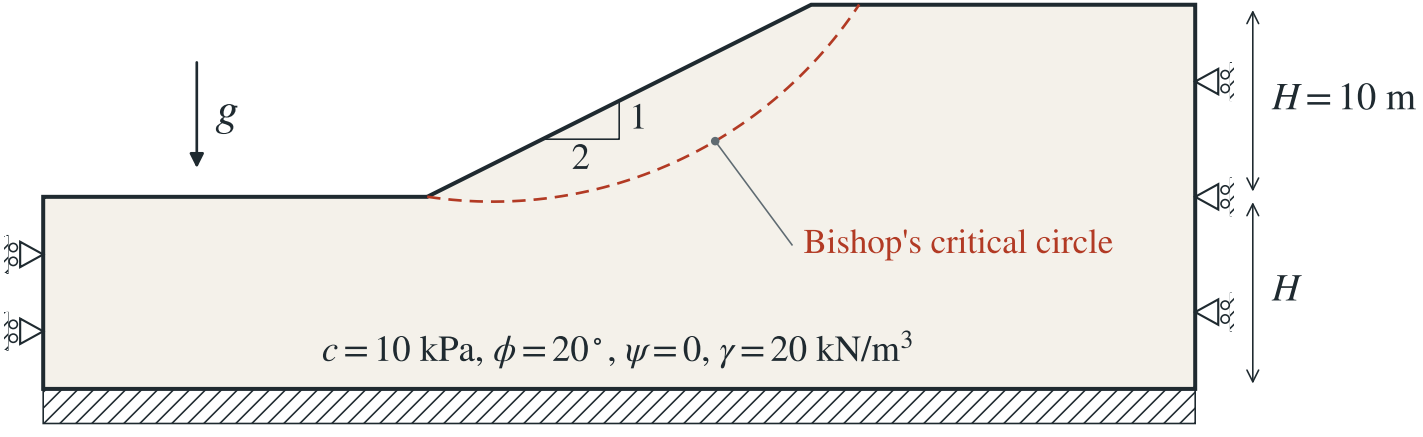
strength reduction: c/F, $\tan\phi/F$; the largest F that stands is FS

base fixed, sides on rollers; gravity applied elastically first